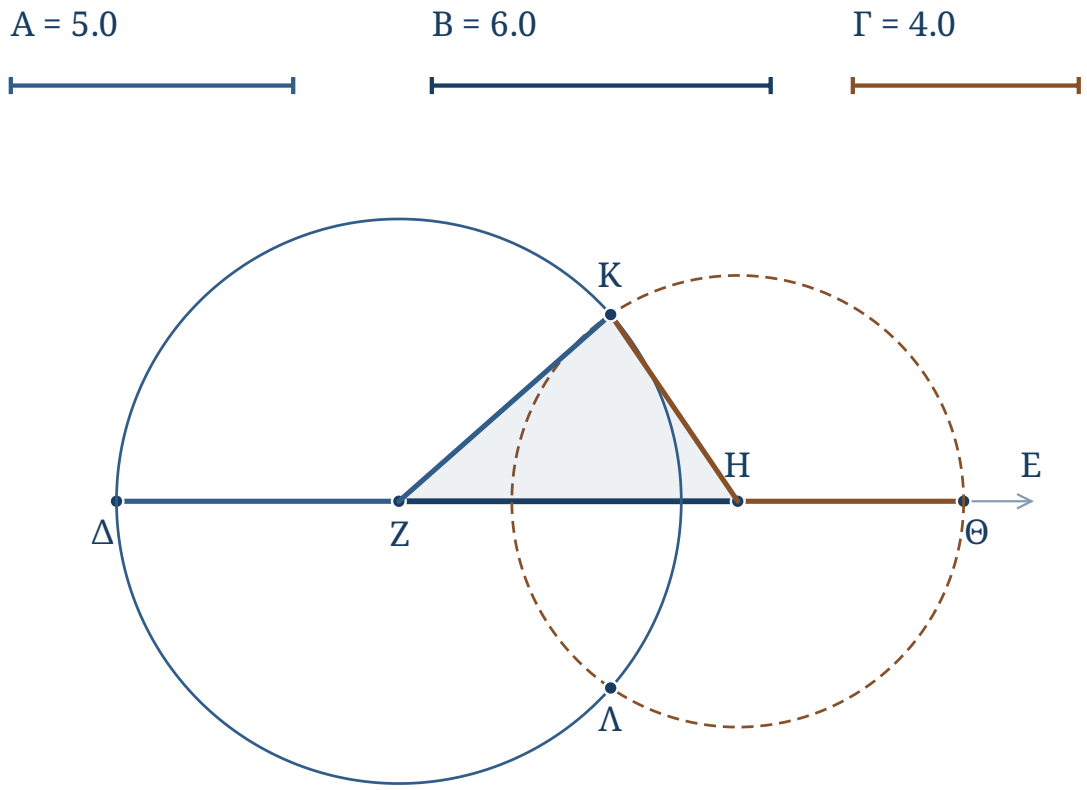
Three lengths. Can they close?

A = 5.0, B = 6.0, Γ = 4.0. Arbitrary drawing units. Construction step 5 of 5. Vertex chosen above the base.



Equal scale on both axes · arbitrary drawing units

The diagram retains the saved view. Its numbers describe a modern coordinate model, not measurements of an ancient instrument.

Two crossings. A triangle on either side.

All three strict inequalities hold. Each crossing supplies the two radii and the base; reflecting the vertex keeps the same three lengths.

Height: **3.307** units. Area: **9.922** square units.

Every pair must be greater

Pair	Your lengths	Strict test
A + B > Γ	11.0 > 4.0	Holds
A + Γ > B	9.0 > 6.0	Holds
B + Γ > A	10.0 > 5.0	Holds

With A and B fixed:

$$1.0 < \Gamma < 11.0$$

The endpoints are excluded.

The whole passage

Greek and Latin: J. L. Heiberg, *Euclidis opera omnia*, vol. I (1883), pp. 52–55, public domain. Latin is Heiberg's nineteenth-century translation. English and teaching notes: Stoicheia, 2026, MIT. The original proof assumes all three strict inequalities; failed experimental givens do not alter the printed text.

---

1. A construction, with a condition

Ἐκ τριῶν εὐθειῶν, αἵ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις [εὐθείαις], τρίγωνον συστήσασθαι· δεῖ δὲ τὰς δύο τῆς λοιπῆς μείζονας εἶναι πάντῃ μεταλαμβανομένας [διὰ τὸ καὶ παντὸς τριγώνου τὰς δύο πλευρὰς τῆς λοιπῆς μείζονας εἶναι πάντῃ μεταλαμβανομένας].

Ex tribus rectis, quae tribus datis aequales sunt, triangulum construere (oportet autem duas reliqua maiores esse quoquo modo coniunctas).

To construct a triangle from three straight lines equal to three given [straight lines]. The two must be greater than the remaining one, however they are taken [because in every triangle, too, two sides are greater than the remaining side, however they are taken].

Language. συστήσασθαι / construere is an infinitive naming the task. δεῖ / oportet introduces a requirement. πάντῃ μεταλαμβανομένας / quoquo modo coniunctas means that every choice of two must be checked. Heiberg brackets the final Greek explanation; the Latin instead cites I.20 in an inline reference, omitted here.

Reason. I.20 proves the requirement for an existing triangle. I.22 now attempts the converse task: construct a triangle from lengths that meet that requirement.

2. Check all three pairings

Ἔστωσαν αἱ δοθεῖσαι τρεῖς εὐθεῖαι αἱ Α, Β, Γ, ὧν αἱ δύο τῆς λοιπῆς μείζονες ἔστωσαν πάντῃ μεταλαμβανόμεναι, αἱ μὲν Α, Β τῆς Γ, αἱ δὲ Α, Γ τῆς Β, καὶ ἔτι αἱ Β, Γ τῆς Α· δεῖ δὴ ἐκ τῶν ἴσων ταῖς Α, Β, Γ τρίγωνον συστήσασθαι.

Sint tres datae rectae A, B, Γ, quarum duae reliqua maiores sint quoquo modo coniunctae, A + B > Γ, A + Γ > B, B + Γ > A. oportet igitur ex rectis aequalibus rectis A, B, Γ triangulum construere.

Let A, B and Γ be the three given straight lines, and let any two be greater than the remaining one: A and B greater than Γ, A and Γ greater than B, and also B and Γ greater than A. We must construct a triangle from lines equal to A, B and Γ.

Language. ἔστωσαν / sint: ‘let them be’, plural. The genitive τῆς Γ follows the comparative μείζονες: ‘greater than Γ’. Here A, B and Γ name whole lengths, not the vertices of a triangle.

Reason. The givens include three strict inequalities. The experiments deliberately allow failures, but Euclid's proof below applies only while all three hold.

---

3. Place the lengths consecutively

Ἐκκείσθω τις εὐθεῖα ἡ ΔΕ πεπερασμένη μὲν κατὰ τὸ Δ ἄπειρος δὲ κατὰ τὸ Ε, καὶ κείσθω τῇ μὲν Α ἴση ἡ ΔΖ, τῇ δὲ Β ἴση ἡ ΖΗ, τῇ δὲ Γ ἴση ἡ ΗΘ.

sumatur recta ΔΕ terminata in Δ, uersus Ε autem infinita, et ponatur ΔΖ = Α, ΖΗ = Β, ΗΘ = Γ.

Set out a straight line ΔΕ, ending at Δ and extending without end towards Ε. Place ΔΖ equal to Α, ΖΗ equal to Β, and ΗΘ equal to Γ.

Language. πεπερασμένη μὲν … ἄπειρος δὲ contrasts an endpoint on one side with no endpoint on the other. κείσθω / ponatur commands the placement. The line on the screen is necessarily only a finite part of that ray.

Reason. The preceding length-transfer and cut-off constructions (I.2–3) supply the three copies. These are reused results, not a new permission to carry an arbitrary compass opening. ΖΗ, the middle copy, becomes the base.

4. Draw the first circle

καὶ κέντρῳ μὲν τῷ Ζ, διαστήματι δὲ τῷ ΖΔ κύκλος γεγράφθω ὁ ΔΚΛ·

et centro Z radio autem ΖΔ circulus describatur ΔΚΛ.

With centre Z and distance ΖΔ, let the circle ΔΚΛ be drawn.

Language. κέντρῳ … διαστήματι are datives specifying centre and distance; Latin uses centro … radio. γεγράφθω is a perfect passive imperative: ‘let it have been drawn’. Κ and Λ name the crossings anticipated in the successful construction.

Reason. Postulate 3 supplies the circle at an available centre through an available point. Every point of this circumference is the required distance A from Z.

---

5. Draw the second; join at K

πάλιν κέντρῳ μὲν τῷ Η, διαστήματι δὲ τῷ ΗΘ κύκλος γεγράφθω ὁ ΚΛΘ, καὶ ἐπεζεύχθωσαν αἱ ΚΖ, ΚΗ· λέγω, ὅτι ἐκ τριῶν εὐθειῶν τῶν ἴσων ταῖς Α, Β, Γ τρίγωνον συνέσταται τὸ ΚΖΗ.

rursus centro H radio autem ΗΘ circulus describatur ΚΛΘ, et ducantur ΚΖ, ΚΗ. dico, ex tribus rectis aequalibus rectis A, B, Γ triangulum constructum esse ΚΖΗ.

Again, with centre H and distance ΗΘ, let the circle ΚΛΘ be drawn, and join ΚΖ and ΚΗ. I say that triangle ΚΖΗ has been constructed from three straight lines equal to A, B and Γ.

Language. πάλιν / rursus repeats the operation with another centre. ἐπεζεύχθωσαν / ducantur is plural: two lines are to be joined. συνέσταται says that the triangle has now been constructed. The interface separates drawing the second circle from joining the vertex.

Reason. Euclid proceeds directly to joining Κ. The existence of the common point needs justification beyond merely drawing two circles. The modern inequality and coordinate explanation below supplies a model, not a missing sentence in the ancient proof.

6. The first radius supplies A

Ἐπεὶ γὰρ τὸ Ζ σημεῖον κέντρον ἐστὶ τοῦ ΔΚΛ κύκλου, ἴση ἐστὶν ἡ ΖΔ τῇ ΖΚ· ἀλλὰ ἡ ΖΔ τῇ Α ἐστιν ἴση· καὶ ἡ ΚΖ ἄρα τῇ Α ἐστιν ἴση.

nam quoniam Z punctum centrum est circuli ΔΚΛ, erit ΖΔ = ΖΚ; uerum ΖΔ = A; quare etiam ΚΖ = A.

Since point Z is the centre of circle ΔΚΛ, ΖΔ equals ΖΚ. But ΖΔ equals A; therefore ΚΖ also equals A.

Language. ἐπεὶ γάρ / nam quoniam begins the reason. ἀλλά / uerum supplies the previously established equality. ἄρα / quare signals the conclusion obtained by connecting them.

Reason. Radii of one circle are equal (definition 15 in this course). Common notion 1 carries the equality through the copied length ΖΔ. A measurement is a check of this drawing, not Euclid's reason.

---

7. The other radius supplies Γ

πάλιν, ἐπεὶ τὸ Η σημεῖον κέντρον ἐστὶ τοῦ ΛΚΘ κύκλου, ἴση ἐστὶν ἡ ΗΘ τῇ ΗΚ· ἀλλὰ ἡ ΗΘ τῇ Γ ἐστιν ἴση· καὶ ἡ ΚΗ ἄρα τῇ Γ ἐστιν ἴση.

rursus quoniam H punctum centrum est circuli ΛΚΘ, erit ΗΘ = ΗΚ; uerum ΗΘ = Γ; quare etiam ΚΗ = Γ.

Again, since point H is the centre of circle ΛΚΘ, ΗΘ equals ΗΚ. But ΗΘ equals Γ; therefore ΚΗ also equals Γ.

Language. The same reasoning returns with H as centre and Γ as the given length. The circle is named ΛΚΘ here, rather than ΚΛΘ as in the construction; the transcription retains that change of order.

Reason. The second circumference certifies the second copied radius. A point on the first circle alone is not enough: this same K must also lie on the circle at H.

8. Keep the middle copy as the base

ἐστὶ δὲ καὶ ἡ ΖΗ τῇ Β ἴση· αἱ τρεῖς ἄρα εὐθεῖαι αἱ ΚΖ, ΖΗ, ΗΚ τρισὶ ταῖς Α, Β, Γ ἴσαι εἰσίν.

et praeterea ZH = B. itaque tres rectae KZ, ZH, HK tribus A, B, Γ aequales sunt.

And ZH also equals B. The three straight lines KZ, ZH and HK therefore equal the three A, B and Γ.

Language. τρεῖς is nominative, the three lines being discussed; τρισί is dative, the three lengths to which they are equal. The Latin likewise contrasts tres and tribus.

Reason. ZH was copied before either circle was drawn. The two circles supply the other two sides; they do not change the base to make the triangle fit.

---

9. What was required to be made

Ἐκ τριῶν ἄρα εὐθειῶν τῶν ΚΖ, ΖΗ, ΗΚ, αἵ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις εὐθείαις ταῖς Α, Β, Γ, τρίγωνον συνέσταται τὸ ΚΖΗ· ὅπερ ἔδει ποιῆσαι.

Ergo ex tribus rectis KZ, ZH, HK, quae tribus datis rectis A, B, Γ aequales sunt, triangulus constructus est KZH; quod oportebat fieri.

Thus triangle KZH has been constructed from the three straight lines KZ, ZH and HK, equal to the three given straight lines A, B and Γ: which was required to be made.

Language. ὅπερ ἔδει ποιῆσαι / quod oportebat fieri closes a construction. Compare ὅπερ ἔδει δεῖξαι, ‘what was required to be shown’, at the end of I.20's theorem.

Reason. The conclusion belongs to the successful case with the original three strict inequalities. At a tangent there is only a straight configuration; when the circles miss, there is no common vertex at all.

Reading this edition

Line breaks, critical apparatus and inline Latin references are omitted. Greek ϑ is normalized to θ; Greek point labels and Heiberg's square brackets are retained. The English translates the Greek, including its bracketed explanation. ΔE is a ray; the drawing shows a finite part. T, P and Q are modern labels. The coordinate method establishes intersection in the real plane; it is not a derivation from Euclid's five postulates.

The printed evidence

Πάλιν, ἐπεὶ παντὸς τριγώνου ἡ ἐκτὸς γωνία τῆς ἐντὸς καὶ ἀπεναντίον μείζων ἐστίν, τοῦ ΓΔΕ ἄρα τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ ΒΔΓ μείζων ἐστὶ τῆς ὑπὸ ΓΕΔ. διὰ ταὐτὰ τοίνυν καὶ τοῦ ΑΒΕ τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ ΓΕΒ μείζων ἐστὶ τῆς ὑπὸ ΒΑΓ. ἀλλὰ τῆς ὑπὸ ΓΕΒ μείζων ἐδείχθη ἡ ὑπὸ ΒΔΓ· πολλῷ ἄρα ἡ ὑπὸ ΒΔΓ μείζων ἐστὶ τῆς ὑπὸ ΒΑΓ.

Ἐὰν ἄρα τριγώνου ἐπὶ μιᾶς τῶν πλευρῶν ἀπὸ τῶν περάτων δύο εὐθεῖαι ἐντὸς συσταθῶσιν, αἱ συσταθεῖσαι τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ἐλάττονες μέν εἰσιν, μείζονα δὲ γωνίαν περιέχουσιν· ὅπερ ἔδει δεῖξαι.

κβ'.

Ἐκ τριῶν εὐθειῶν, αἵ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις [εὐθείαις], τρίγωνον συστήσασθαι· δεῖ δὲ τὰς δύο τῆς λοιπῆς μείζονας εἶναι πάντῃ μεταλαμβανομένας [διὰ τὸ καὶ παντὸς τριγώνου τὰς δύο πλευρὰς τῆς λοιπῆς μείζονας εἶναι πάντῃ μεταλαμβανομένας].

Ἔστωσαν αἱ δοθεῖσαι τρεῖς εὐθεῖαι αἱ Α, Β, Γ, ὧν αἱ δύο τῆς λοιπῆς μείζονες ἔστωσαν πάντῃ μεταλαμβανόμεναι, αἱ μὲν Α, Β τῆς Γ, αἱ δὲ Α, Γ τῆς Β, καὶ ἔτι αἱ Β, Γ τῆς Α· δεῖ δὴ ἐκ τῶν ἴσων ταῖς Α, Β, Γ τρίγωνον συστήσασθαι.

Ἐκκείσθω τις εὐθεῖα ἡ ΔΕ πεπερασμένη μὲν κατὰ

XXII. Proclus p. 102, 16. Eutocius in Apollonium p. 10. Boetius p. 382, 1 (male). partem demonstrationis habet Proclus p. 330 sq.

---

2. ἐντός] ἐν- in ras. b. ἐστίν] PF; ἐστί uulgo. ΓΔΕ] e corr. F m. 2; mutat. in ΓΕΔ V. ἄρα] supra F. 3.

Heiberg 1883, printed page 53 · Latin

rursus quoniam in quouis triangulo angulus extrinsecus positus maior est angulo interiore et opposito [prop. XVI], in triangulo $\Gamma\Delta E$ erit $\angle B\Delta\Gamma > \Gamma E\Delta$. eadem de causa igitur etiam in triangulo ABE erit $\angle\Gamma EB > BA\Gamma$. uerum demonstratum est $\angle B\Delta\Gamma > \Gamma EB$. multo igitur magis $B\Delta\Gamma > BA\Gamma$.

Ergo si in uno latere trianguli a terminis duae rectae intus coniunguntur, rectae coniunctae reliquis duobus lateribus trianguli minores erunt, maiorem autem angulum comprehendent; quod erat demonstrandum.

XXII.

Ex tribus rectis, quae tribus datis aequales sunt, triangulum construere (oportet autem duas reliqua maiores esse quoquo modo coniunctas [prop. XX]).

Sint tres datae rectae A, B, Γ, quarum duae reliqua maiores sint quoquo modo coniunctae, $A + B > \Gamma$, $A + \Gamma > B$, $B + \Gamma > A$. oportet igitur ex rectis aequalibus rectis A, B, Γ triangulum construere.

sumatur[1]) recta ΔE terminata in Δ, uersus E au-

1) Proclum non ipsa uerba Euclidis citare, adparet. cfr. idem p. 102, 19. Augustum perperam post $K\Lambda\Theta$ p. 54, 5. suppleuisse: *καὶ τεμνέτωσαν ἀλλήλους οἱ κύκλοι κατὰ τὸ K*, demonstraui „Studien“ p. 185.

$B\Delta\Gamma$] Δ in ras. F. *ἐστιν* PV. 4. $\Gamma E\Delta$] eras. F. *ταὐτά*] *τὰ αὐτά* F; *ταῦτα* Vbp. 5. *ἐστίν* P, ut lin. 7. 6. *ἀλλὰ καὶ τῆς* F. 7. $B\Delta\Gamma$] (alt.) $B\Delta$ in ras. sunt V. 12. *εἰσιν*] P; *εἰσι* uulgo. 15. *αἳ εἰσιν τρισὶ ταῖς δοθείσαις εὐθείαις ἴσαι* Proclus p. 329; sed p. 102: *αἳ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις εὐθείαις.* 16. *εὐθείαις*] om. b; m. rec. P; supra p; mg. m. 2 V; om. Eutocius. 17. *δέ*] Proclus, Eutocius; *δή* codd. *τάς*] corr. ex *ταῖς* F. *δύο*] *β* b. 18. *διὰ τὸ* — 20. *μεταλαμβανομένας*] omnes codd., Boetius; om. Proclus, Campanus; contra Eutocius ea habuisse uidetur. 21. *τρεῖς*] om. p.

Heiberg 1883, printed page 54 · Greek

τὸ Δ ἄπειρος δὲ κατὰ τὸ Ε, καὶ κείσθω τῇ μὲν Α ἴση ἡ ΔΖ, τῇ δὲ Β ἴση ἡ ΖΗ, τῇ δὲ Γ ἴση ἡ ΗΘ· καὶ κέντρῳ μὲν τῷ Ζ, διαστήματι δὲ τῷ ΖΔ κύκλος γεγράφθω ὁ ΔΚΛ· πάλιν κέντρῳ μὲν τῷ Η, διαστήματι δὲ τῷ ΗΘ κύκλος γεγράφθω ὁ ΚΛΘ, καὶ ἐπεζεύχθωσαν αἱ ΚΖ, ΚΗ· λέγω, ὅτι ἐκ τριῶν εὐθειῶν τῶν ἴσων ταῖς Α, Β, Γ τρίγωνον συνέσταται τὸ ΚΖΗ.

Ἐπεὶ γὰρ τὸ Ζ σημεῖον κέντρον ἐστὶ τοῦ ΔΚΛ κύκλου, ἴση ἐστὶν ἡ ΖΔ τῇ ΖΚ· ἀλλὰ ἡ ΖΔ τῇ Α ἐστιν ἴση. καὶ ἡ ΚΖ ἄρα τῇ Α ἐστιν ἴση. πάλιν, ἐπεὶ τὸ Η σημεῖον κέντρον ἐστὶ τοῦ ΛΚΘ κύκλου, ἴση ἐστὶν ἡ ΗΘ τῇ ΗΚ· ἀλλὰ ἡ ΗΘ τῇ Γ ἐστιν ἴση· καὶ ἡ ΚΗ ἄρα τῇ Γ ἐστιν ἴση. ἐστὶ δὲ καὶ ἡ ΖΗ τῇ Β ἴση· αἱ τρεῖς ἄρα εὐθεῖαι αἱ ΚΖ, ΖΗ, ΗΚ τρισὶ ταῖς Α, Β, Γ ἴσαι εἰσίν.

Ἐκ τριῶν ἄρα εὐθειῶν τῶν ΚΖ, ΖΗ, ΗΚ, αἵ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις εὐθείαις ταῖς Α, Β, Γ, τρίγωνον συνέσταται τὸ ΚΖΗ· ὅπερ ἔδει ποιῆσαι.

κγ΄.

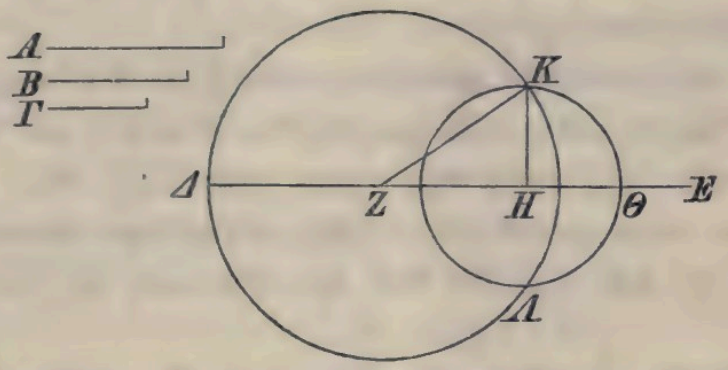
Πρὸς τῇ δοθείσῃ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῇ δοθείσῃ γωνίᾳ εὐθυγράμμῳ ἴσην γωνίαν εὐθύγραμμον συστήσασθαι.

XXIII. Boetius p. 382, 5.

---

1. τῇ] postea insertum m. 1 V. 2. ἡ] (tert.) m. rec. P. 3. μέν] om. b, Proclus. 4. καὶ πάλιν V, Proclus. μέν] om. V, Proclus. διαστήματι δέ] καὶ διαστήματι P. 7. συνέστηκε V; συνίσταται p. τό] corr. ex τῷ b. 8. γάρ] οὖν P. ἐστίν P. 9. ΖΔ] ΔΖ F. ἀλλ F. ΖΔ] ΔΖ V (ante Δ ras., Ζ mg. m. 2). 10. καὶ ἡ ΚΖ ἄρα τῇ Α ἐστιν ἴση] mg. m. 2 V. 11. ἐστίν Bb. ΛΚΘ] ΚΛΘ P, et in ras. V. 12. ἀλλ' F. 13. ΚΗ] corr. ex ΚΘ m. 2 P. 14. ΗΚ BF. ἐστιν ἴση] mg. m. 2 V. ἐστὶν δέ P. 16. τῶν]

tem infinita, et ponatur $\Delta Z = A$, $ZH = B$, $H\Theta = \Gamma$. et centro Z radio autem $Z\Delta$ circulus describatur $\Delta K\Lambda$. rursus centro H radio autem $H\Theta$ circulus describatur $K\Lambda\Theta$, et ducantur KZ, KH. dico, ex tribus rectis aequalibus rectis A, B, Γ triangulum constructum esse KZH.



nam quoniam Z punctum centrum est circuli $\Delta K\Lambda$, erit $Z\Delta = ZK$; uerum $Z\Delta = A$; quare etiam $KZ = A$ [κ. ἔννν. 1].[1]) rursus quoniam H punctum centrum est circuli $\Delta K\Theta$, erit $H\Theta = HK$; uerum $H\Theta = \Gamma$; quare etiam $KH = \Gamma$. et praeterea $ZH = B$. itaque tres rectae KZ, ZH, HK tribus A, B, Γ aequales sunt.

Ergo ex tribus rectis KZ, ZH, HK, quae tribus datis rectis A, B, Γ aequales sunt, triangulus constructus est KZH; quod oportebat fieri.

XXIII.

Ad datam rectam et punctum in ea datum angulum rectilineum dato angulo rectilineo aequalem construere.

1) Cfr. Alexander Aphrod. in anal. I fol. 8. Studien p. 195.

τοῦ F. 17. τρισί] om. F. Γ] om. V. 18. συνίσταται p. 21. εὐθυγράμμῳ γωνίᾳ Proclus.

Reopen these givens in the workshop · Full source volume · Code and limitations

This file contains its font, diagram, complete reading and four scans. Opening it makes no external requests. Print with your browser's print command.